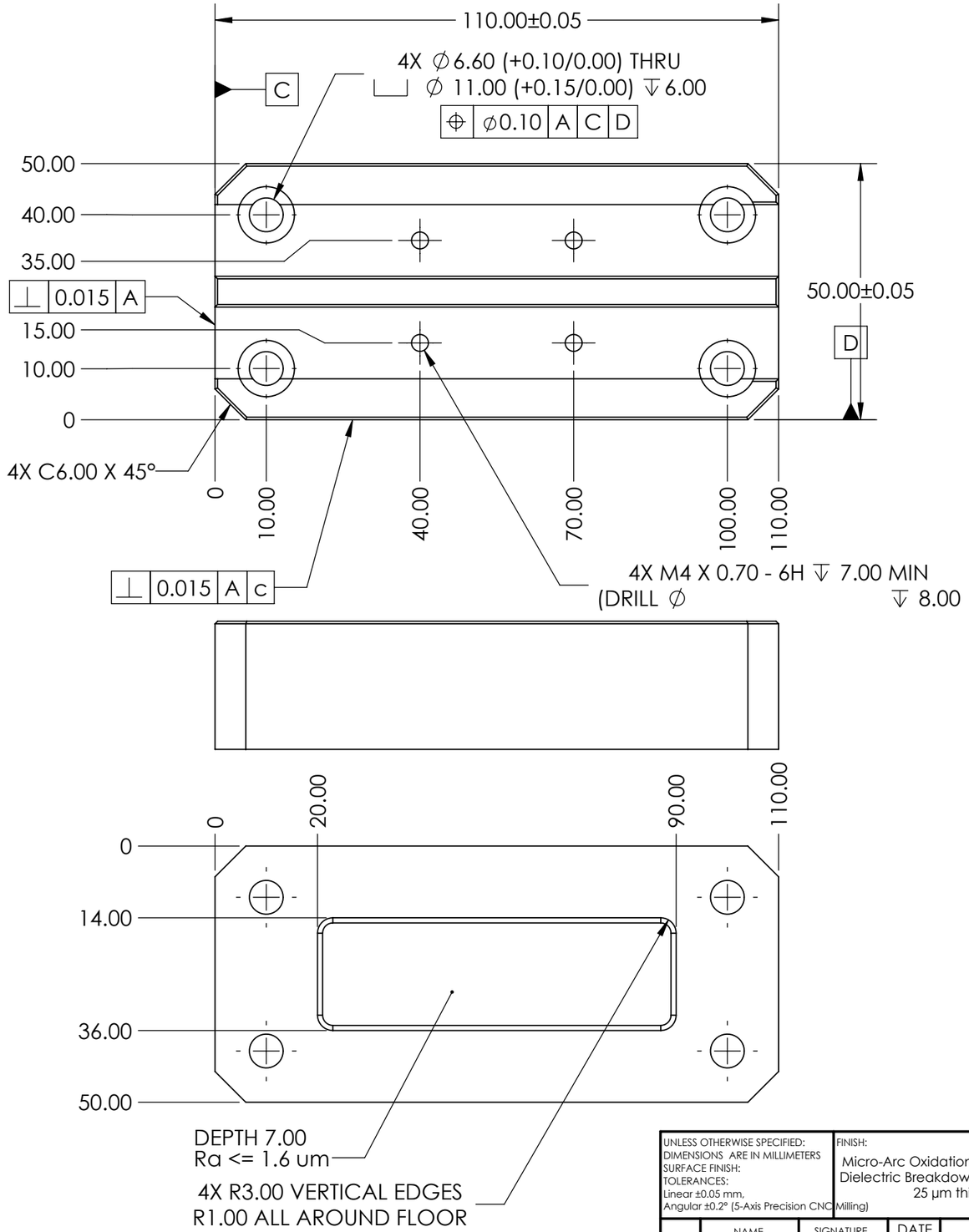
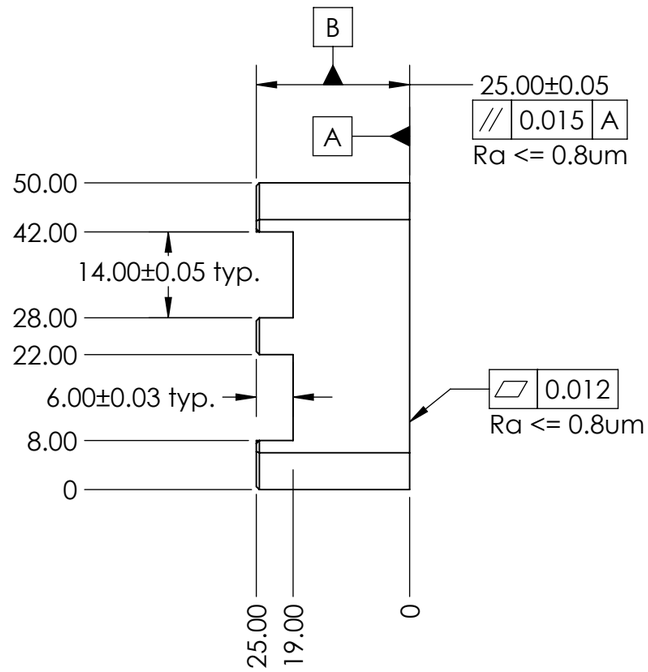
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-EV-BUSBAR-BRACKET-4.3.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/-0.2 DEG (5-AXIS PRECISION CNC MILLING).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
   1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. MATERIAL & ELECTRICAL INSULATION ASSURANCE:
   2.1 MATERIAL: FORGED AEROSPACE/EV ALUMINUM ALLOY 6082-T6 (EN AW-6082 / ASTM B221), THERMAL CONDUCTIVITY k >= 170 W/(m·K).
   2.2 DIELECTRIC SURFACE COATING: HIGH-PERFORMANCE MICRO-ARC OXIDATION (MAO / PEO, 25 +/-3 MICRONS THICK).
   2.3 ELECTRICAL BREAKDOWN STRENGTH: WITHSTAND >= 3000V DC HI-POT TEST FOR 60 SECONDS (LEAKAGE CURRENT < 0.5 mA) ACROSS ALL BUSBAR CONTACT SURFACES (IEC 60664-1 COMPLIANT).

3. LIQUID COOLING POCKET & GD&T CONTROLS:
   3.1 PRIMARY DATUM [A]: FLATNESS <= 0.012 mm, SURFACE FINISH Ra <= 0.8 MICRONS FOR COLD PLATE SEALING.
   3.2 BUSBAR MATING SURFACES [B]: PARALLEL TO [A] WITHIN 0.015 mm, Ra <= 0.8 MICRONS.
   3.3 BOTTOM COOLING POCKET: DEPTH 7.00 mm, 4X VERTICAL CORNER FILLET R3.00 mm (MACHINED VIA Ø6 mm ENDMILL), FLOOR PERIMETER FILLET R1.00 mm TO MINIMIZE STRESS CONCENTRATION AND FLOW DRAG.
   3.4 100% SUBTRACTIVE CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING PERMITTED (IATF 16949 COMPLIANT).





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.2° (5-Axis Precision CNC Milling)	FINISH: Micro-Arc Oxidation (MAO / PEO, Dielectric Breakdown > 3000V DC, 25 µm thick)	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION: A

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE: 800V EV FAST-CHARGING INTEGRATED BUSBAR THERMAL BRACKET

MATERIAL: Forged Aluminum Alloy 6082-T6 (EN AW-6082) / Al 6061-T6 (AMS 4027)

DWG NO. W3-EV-BUSBAR-BRACKET-4.3

A3

WEIGHT: ~273.7 g

SCALE NTS

SHEET 1 OF 1